Enterprise AI Architecture & Infrastructure Strategy (2026)

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1. Executive Summary

This strategy document defines the engineering framework for deploying scalable, fault-tolerant Retrieval-Augmented Generation (RAG) and Agentic AI systems across our global cloud infrastructure. By standardizing on hybrid embedding pipelines, distributed vector indexes, and deterministic guardrail policies, the platform achieves sub-250ms token time-to-first-token (TTFT) and maintains a 99.95% service availability SLA.

2. Strategic Architectural Objectives

- **Latency Targets:** P95 end-to-end response latency under 650ms for document retrieval queries up to 50 pages.
- **Cost Efficiency:** 42% reduction in inference operational costs achieved through selective model routing (Flash models for classification and retrieval summarization; Pro models reserved for complex multi-hop synthesis).
- **Data Sovereignty & Compliance:** Full adherence to GDPR, SOC2 Type II, and HIPAA standards with zero data retention policies on model provider endpoints.
- **Context Window Optimization:** Utilization of token-aware hierarchical chunking with a 500-token window and 10% overlap to maximize cosine retrieval precision.

3. Vector Storage and Retrieval Subsystem

The core indexing infrastructure employs high-dimensional dense embeddings (3072 dimensions) with Cosine similarity space. Vector search is augmented by a hybrid sparse-dense reciprocal rank fusion (RRF) algorithm to mitigate vocabulary mismatch issues.

3.1 Benchmarks & Model Evaluation

During Q1 internal benchmarking across 100,000 technical documents, our evaluation pipeline recorded the following metrics: Mean Reciprocal Rank (MRR@5) improved from 0.74 to 0.91 when switching from fixed-length character splitting to token-aware recursive boundaries. Context hallucination rates dropped from 6.8% to under 0.4% with the enforcement of strict negative-constraint system instructions.

4. Security, Governance & Guardrails

All incoming user prompts and uploaded documents pass through an automated four-stage governance filter before LLM processing:

1. **Payload Validation:** Whitelist file validation and binary header integrity checking up to 20MB file limit.
2. **PII & Secret Sanitization:** Automated regex and Named Entity Recognition (NER) masking for API keys, Social Security numbers, and credit card credentials.
3. **Prompt Injection Shield:** Heuristic semantic classifiers detecting jailbreak attempts and system prompt overrides.
4. **Attribution Verification:** Real-time citation validator verifying that all synthesized claims correspond to verified source chunk spans with >= 70% confidence.

5. Production Deployment & SLA Framework

The production rollout is orchestrated via Kubernetes container clusters across multi-region availability zones. Auto-scaling policies trigger horizontal pod expansion whenever request concurrency exceeds 85 requests per second per instance. Continuous telemetry is monitored via distributed OpenTelemetry tracing and structured JSON logging.